AI-Assisted Pharmaceutical Document Change Analyzer

1. Project Objective

Build a small web application that compares two versions of a pharmaceutical document and identifies:

- Added content
- Deleted content
- Modified content
- Numeric changes
- Specification changes
- Date and reference changes
- Potentially critical changes requiring review

The application should produce a concise change report with source evidence.

2. Business Use Case

When an SOP, specification, analytical method, protocol, report, or product-development document is revised, reviewers must manually compare the old and new versions.

Simple text-difference tools often fail because:

- Paragraphs may move between sections
- Tables may change
- Formatting changes create unnecessary noise
- Small numeric changes may be critical
- Regulatory or procedural meaning may change even when wording changes only slightly

This project should help a reviewer quickly understand what changed and which changes may be important.

3. Example Input

Version 1

Assay acceptance criterion: 95.0% to 105.0%.

Samples shall be stored at 25°C ± 2°C.

The Quality Control Manager shall approve the result.

Version 2

Assay acceptance criterion: 98.0% to 102.0%.

Samples shall be stored at 25°C ± 2°C and 60% RH ± 5% RH.

The Quality Assurance Manager shall approve the result.

Expected Output

Change	Category	Risk	Reason
Assay range changed from 95.0–105.0% to 98.0–102.0%	Specification	High	Acceptance criterion became narrower
Added 60% RH ± 5% RH storage condition	Process condition	High	Additional environmental requirement
QC Manager changed to QA Manager	Responsibility	Medium	Approval responsibility changed

4. Supported Documents

For the MVP, support:

- Text-based PDF
- DOCX
- TXT
- Scanned document (phase 2)

5. Core Features

A. Upload Two Versions

The user should upload:

- Previous document
- Revised document

The application should display the filenames and extracted text.

B. Text Extraction

Extract text from:

- PDF using PyMuPDF
- DOCX using python-docx
- TXT using normal file reading

Retain page numbers or paragraph numbers wherever possible.

C. Section Matching

The system should attempt to match corresponding sections even when:

- Section numbers change
- Paragraphs move
- Headings are slightly renamed
- Additional sections are inserted

Example:

Version 1: 5.2 Sample Preparation

Version 2: 6.1 Preparation of Sample Solution
These sections should be identified as potentially corresponding sections.

D. Change Detection

Identify:

- Added paragraph
- Deleted paragraph
- Modified paragraph
- Moved paragraph
- Numeric change
- Unit change
- Date change
- Role or responsibility change
- Reference-document change
- Specification change

E. Risk Classification

Classify each change as:

- High
- Medium
- Low
- Informational

Suggested rules:

Change type	Suggested risk
Acceptance criterion or specification changed	High
Temperature, time, speed, pressure or quantity changed	High
Process sequence changed	High
Approval role changed	Medium
Regulatory reference changed	Medium
Test method reference changed	Medium
Additional clarification without meaning change	Low
Formatting, spelling or punctuation only	Informational

The intern may use an LLM to assist classification, but important numeric changes must also be detected through deterministic code.

F. Change Report

The output should contain:

- Document names
- Change summary
- Number of high-, medium- and low-risk changes
- Old text
- New text
- Change category
- Risk level
- Explanation
- Page or paragraph reference

6. Expected Output Schema

```
{
 "comparison_id": "CMP-001",
 "old_document": "SOP_v1.pdf",
 "new_document": "SOP_v2.pdf",
 "summary": {
  "total_changes": 12,
  "high_risk": 3,
  "medium_risk": 4,
  "low_risk": 3,
  "informational": 2
 },
 "changes": [
  {
   "change_id": "CH-001",
   "section": "Acceptance Criteria",
   "change_type": "numeric_specification_change",
   "old_text": "Assay: 95.0% to 105.0%",
   "new_text": "Assay: 98.0% to 102.0%",
   "risk_level": "HIGH",
   "reason": "The approved assay acceptance range was narrowed.",
   "old_page": 4,
   "new_page": 5,
   "confidence": 0.97
  }
 ]
}
```

7. Suggested Technology

- **Frontend:** Streamlit
- **Backend:** Fastapi

- **Optional LLM:** Gemini (For now)
- **Database:** SQLite
- **Packaging:** Docker

8. Recommended Application Pages

Page 1: Upload and Compare

- Upload old document
- Upload revised document
- Start comparison
- Show processing status

Page 2: Change Summary

Show:

- Total changes
- High-risk changes
- Medium-risk changes
- Low-risk changes
- Informational changes

Page 3: Detailed Changes

Display a table containing:

Section	Old text	New text	Change type	Risk	Reason

Allow filtering by:

- Risk
- Section
- Change type

Page 4: Review and Export

Allow the reviewer to:

- Confirm risk classification
- Modify risk level
- Add a reviewer comment
- Mark a change as accepted

- Export the report as CSV or JSON